Design and Analysis of a Miller-Compensated Two-Stage Op-Amp

Project Title

Design and Analysis of a Miller-Compensated Two-Stage Op-Amp

Objectives of the Project

- To design a two-stage CMOS operational amplifier in 90nm CMOS technology, consisting of a differential input stage and a common-source gain stage.
- To achieve high open-loop DC gain with relatively good bandwidth and phase margin at low bias currents.
- To characterize the amplifier's differential-mode gain, common-mode gain, phase margin, slew rate, gain-bandwidth, and power dissipation through simulation.
- To validate the schematic-level design and extract performance metrics using Cadence Virtuoso.

Design Specifications

Parameter	**Target Value**
Technology	90nm CMOS (gpdk090)
Supply Voltage (VDD)	1.2 V (0 V to 1.2 V swing)
DC Gain (Av)	52 dB
Phase Margin	60°
Slew Rate	5 V/µs
Power Dissipation	1 mW
Load Capacitance (CL)	2 pF
CMRR	High
Bias Current	Low bias current design

Design Methodology

- Topology selection – A two-stage architecture is chosen: a PMOS/NMOS differential pair as the first stage followed by a common-source second stage.
- Calculations – Initial sizing of transistors (W/L ratios) is derived using MOSFET square-law equations from the target gain and slew rate specifications, using 90nm process parameters.
- Compensation network design – A Miller capacitor (Cc) is placed between the first and second stage outputs to split the dominant and non-dominant poles.
- Schematic entry and simulation – The circuit is entered and simulated in Cadence Virtuoso to verify DC operating points, open-loop gain/phase and slew rate.
- Iterative tuning – Transistor sizes, bias currents and Cc are iteratively adjusted to meet all specifications simultaneously.
- Corner and parametric analysis – Gain, phase margin, slew rate and power dissipation are verified across process/voltage/temperature corners and bias sweeps to confirm robustness.
- Result extraction – Final DC gain, phase margin, slew rate, and power dissipation are extracted from AC and transient simulation results in Cadence Virtuoso.

EDA Tools / Software to be Used

- Cadence Virtuoso – schematic capture and simulation environment
- ADE-L – DC, AC, and transient simulation
- CMOS technology (gpdk090)

Abstract

An operational amplifier is considered to be the fundamental block in mixed signal and analog design. This project aims to design a two stage CMOS operational amplifier using Miller Compensation technique, mainly to achieve high gain with relatively good bandwidth and phase margin with low bias currents in 90 nm technology. Keeping this as a main aspect, opamp specifications are taken into account, i.e., differential mode gain, common mode gain, phase margin, slew rate, power dissipation and gain bandwidth. This design operates at 1.2V to 0V half voltage supply swing. The simulation process is carried out in Cadence Virtuoso in 90nm technology. The obtained DC gain is 52 dB with a phase margin of 60 degrees and a slew rate of 5 V/μSec with a power dissipation of 1 mW.